

Statistical Inference for Regression

1. Point Estimators vs Interval Estimators

A point estimator provides a single best estimate of a population parameter.

Example: Estimated mean salary = \$70,000.

Limitation: It does not reflect uncertainty.

An interval estimator provides a range of plausible values.

Example: Salary is between \$65,000 and \$75,000.

2. Confidence Intervals (CI) and α

Confidence intervals quantify uncertainty around estimates.

- Confidence Level = $1 - \alpha$

Example: $\alpha = 0.05 \rightarrow 95\%$ CI

Interpretation:

If repeated many times, 95% of such intervals would contain the true value.

3. Hypothesis Testing

Used to decide whether an observed effect is statistically significant.

- H_0 : No effect ($\beta = 0$)
- H_1 : Effect exists ($\beta \neq 0$)

We assume H_0 is true and evaluate whether the data is consistent with that assumption.

4. Test Statistic – What It Means

A test statistic is a standardized measure of how far the observed estimate is from the null hypothesis value.

Conceptually:

It answers: “How many standard errors away is my estimate from 0?”

Example:

If coefficient = 2.5 and standard error = 0.5 $\rightarrow$ test statistic = 5.

- Large test statistic $\rightarrow$ strong evidence against H_0
- Small test statistic $\rightarrow$ weak evidence against H_0

The test statistic is what connects raw data to probabilities (p-values).

5. Type I and Type II Errors

Type I Error: Reject H_0 when it is true (false positive).

Type II Error: Fail to reject H_0 when it is false (false negative).

Key question:

If H_0 is true, can we make a Type II error?

Answer: No.

6. Significance Level (α)

α is the probability of making a Type I error.

- $\alpha = P(\text{Type I error})$

Decision rule:

- If $p\text{-value} \leq \alpha \rightarrow \text{Reject } H_0$
- If $p\text{-value} > \alpha \rightarrow \text{Fail to reject } H_0$

7. Rejection Region

The rejection region is defined so that the probability of rejecting H_0 when it is true equals α .

- $P(\text{reject } H_0 \mid H_0 \text{ is true}) = \alpha$

8. P-values

The p-value is the probability of observing a test statistic as extreme as the one calculated, assuming H_0 is true.

- Small p-value $\rightarrow$ reject H_0
- Large p-value $\rightarrow$ fail to reject H_0

Important:

P-value = $P(\text{Data} \mid H_0 \text{ is true})$, not $P(H_0 \text{ is true})$.

9. Connection to Regression

Each regression coefficient includes:

- Point estimate
- Confidence interval
- Test statistic
- P-value
- Decision using α

Final takeaway:

All components work together to determine whether a relationship is real or due to random variation.